

PULLMAN-MOSCOW, WA, USA

WMO: 727857

Lat: 46.744N

Lon: 117.109W

Elev: 2551

StdP: 13.39

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94129

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	7.1	14.0	0.2	6.1	9.5	7.2	8.7	16.4	31.1	38.9	26.7	37.7	4.0	90	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.5	92.4	63.0	89.6	62.4	85.7	61.1	67.0	84.9	64.5	83.1	62.6	81.4	5.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.6	86.8	74.7	57.2	76.7	71.5	55.0	70.8	69.2	32.8	84.7	30.8	82.6	29.4	81.8	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.2	20.4	18.2	DB	-3.1	98.3	8.5	3.1	-9.2	100.5	-14.2	102.3	-18.9	104.0	-25.1	106.2	
			WB	-4.0	69.5	8.2	3.9	-9.9	72.3	-14.6	74.7	-19.2	76.9	-25.1	79.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	32.4	34.4	40.6	46.3	53.7	59.4	67.2	66.8	59.3	48.0	38.4	31.0
	DBStd	14.48	8.58	8.11	6.66	6.41	7.10	6.48	5.88	6.25	7.10	7.34	8.10	8.19
	HDD50	2548	546	437	297	147	41	5	0	0	8	123	356	588
	HDD65	6403	1011	856	757	560	355	190	45	52	198	527	799	1053
	CDD50	1896	0	0	5	37	157	287	534	522	286	61	7	0
	CDD65	281	0	0	0	0	7	22	114	110	27	1	0	0
	CDH74	5654	0	0	0	13	143	465	2234	2136	641	22	0	0
	CDH80	2308	0	0	0	1	28	137	979	938	222	3	0	0
	WSAvg	7.2	8.8	8.4	8.6	8.2	6.8	6.0	4.6	4.9	5.4	6.7	8.6	8.8
	PrecAvg	20.4	2.6	2.0	2.0	1.7	1.7	1.3	0.7	0.8	0.9	1.5	2.8	2.7
Precipitation	PrecMax	28.0	6.2	5.3	4.7	4.6	3.3	3.5	1.7	3.8	2.8	3.3	5.7	6.2
	PrecMin	14.9	0.4	0.1	0.4	0.2	0.4	0.2	0.0	0.0	0.0	0.0	0.5	0.6
	PrecStd	3.4	1.3	1.2	0.9	0.9	0.7	0.7	0.5	0.8	0.7	0.9	1.2	1.3
	PrecStd	3.4	1.3	1.2	0.9	0.9	0.7	0.7	0.5	0.8	0.7	0.9	1.2	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.1	55.0	65.1	75.4	83.3	90.2	96.6	97.1	91.4	77.1	61.9	50.6
		MCWB	45.8	46.7	50.0	56.3	60.3	65.7	64.8	63.5	61.3	57.8	52.2	45.4
	2%	DB	47.0	50.2	58.6	68.5	78.2	83.6	92.4	92.8	86.1	70.2	55.9	46.4
		MCWB	42.3	44.3	47.3	52.6	58.8	63.9	63.9	62.2	60.3	53.7	48.8	42.3
	5%	DB	44.6	47.2	54.6	63.6	73.1	79.2	89.7	89.7	81.5	64.8	52.3	43.5
		MCWB	40.6	42.3	45.8	50.4	56.8	61.1	63.1	61.6	58.3	51.5	46.8	40.0
	10%	DB	41.9	44.6	51.3	58.6	68.7	74.9	85.8	85.4	76.7	60.5	49.0	40.9
		MCWB	38.6	40.1	44.3	47.8	54.2	59.0	61.8	60.2	56.7	49.5	44.2	37.7
	0.4%	WB	46.7	48.3	51.4	58.3	62.8	69.2	71.0	68.6	67.5	59.5	54.3	46.1
		MCDB	50.0	52.4	60.5	71.7	78.8	83.7	89.3	89.9	82.9	74.1	60.0	49.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.9	45.1	49.1	53.9	60.1	65.2	67.5	64.4	62.2	55.3	49.8	43.0
		MCDB	46.3	49.1	56.8	65.9	75.1	81.2	85.1	87.1	81.4	66.3	55.1	46.0
	5%	WB	40.8	42.8	46.8	51.0	57.9	62.3	65.3	62.6	59.8	52.8	47.4	40.5
		MCDB	44.2	46.7	53.3	61.8	70.9	76.8	84.0	84.9	78.2	62.6	51.5	43.7
	10%	WB	38.9	40.4	44.5	48.7	55.6	60.0	63.3	61.1	57.4	50.4	44.8	38.2
		MCDB	42.0	44.1	50.5	57.8	66.3	73.1	81.6	82.5	74.1	59.1	48.5	40.8
Mean Daily Temperature Range	5% DB	MDBR	9.6	12.4	15.8	18.6	21.4	23.6	31.5	31.5	27.7	20.1	12.1	9.7
		MCDBR	11.3	15.4	21.3	26.7	29.5	32.3	37.8	38.3	36.2	27.8	15.6	11.4
		MCWBR	8.4	10.6	13.3	14.5	14.5	16.1	15.5	15.6	17.1	15.3	10.0	8.9
	5% WB	MCDBR	10.7	14.1	19.2	24.3	25.8	30.3	33.1	33.7	32.7	24.1	14.2	10.9
		MCWBR	8.3	10.3	12.9	13.9	13.6	16.1	15.6	14.9	17.0	14.2	9.8	8.8
Clear-Sky Solar Irradiance	taub		0.273	0.276	0.290	0.309	0.329	0.327	0.331	0.338	0.318	0.303	0.277	0.271
	taud		2.520	2.560	2.564	2.497	2.454	2.482	2.468	2.438	2.474	2.529	2.560	2.507
	Ebn at Noon		267	291	300	302	297	297	294	288	285	274	264	254
	Edh at Noon		20	23	27	31	34	34	34	33	29	24	19	18
All-Sky Solar Radiation	RadAvg		376	663	1012	1492	1881	2096	2375	1959	1422	867	440	295
	RadStd		34	78	103	132	148	159	110	128	118	82	39	32

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-3.96	N/A	N/A	N/A	N/A	N/A	+218	N/A
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	N/A

Nomenclature: See separate page